

Samridh Prabhakar

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EDUCATION

Purdue University

B.S. Computer Engineering, Certificate in Semiconductors and Microelectronics

Graduating 2028

GPA: 3.4

EXPERIENCES

Research Assistant

Appenzeller Chen Group, Birck Nanotechnology Center

Aug 2026 – Present

West Lafayette, IN

- Contributed to an **NSF-funded 19M research program** developing automated and remotely accessible ALD infrastructure for next-generation semiconductor device research.
- Working toward fabrication of **transition-metal dichalcogenide (TMD)-based complementary FET (CFET)** structures using 2D semiconductor materials and cleanroom processing.

Undergraduate Researcher

Purdue Vertically Integrated Projects, Semiconductors @ Birck

Jan 2026 – Present

West Lafayette, IN

- Developing a repeatable **deep reactive ion etch (DRIE)** process for **through-silicon vias (TSVs)** as part of an Advanced Packaging and Heterogeneous Integration research team.
- Working with **DRIE and ALD process development** to improve repeatability and integration of wafer-level TSV structures for heterogeneous semiconductor packaging.
- Fabricated 6 wafers of Ti/Al-HfO₂-Si **MIM capacitors** through a full photolithography process flow, targeting 1800 nm AZ1518 resist thickness and 0.28 mm² electrode geometry.
- Developed **statistical process control (SPC) charts** for AZ1518 photoresist thickness and capacitor electrode area across a six-wafer fabrication run.

Project Manager, TARS Robotics Team

Embedded Systems @ Purdue

Aug 2026 – Present

West Lafayette, IN

- Leading a **15-member multidisciplinary team** developing a TARS-inspired mobile robot, coordinating mechanical, electrical, embedded, and software work through milestones, and system integration.

PROJECTS

RP2354 FIDO2 Hardware Security Key | *RP2350, KiCad, USB, C/C++*

- Designed and manufactured a custom **RP2354-based hardware security key PCB** integrating USB connectivity, voltage regulation, ESD protection, oscillator circuitry, and user-presence input.
- Integrated the custom hardware platform with an existing **FIDO2 firmware stack**, adapting firmware to the RP2354 hardware design rather than implementing the authentication protocol from scratch.

FlowGuard | *Python, LibreLane, OpenROAD, SKY130, TinyTapeout*

- Built an **AI-assisted physical-design optimizer** that learns from successful and failed RTL-to-GDS runs to automatically select chip placement, routing, and timing configurations.
- Automated and analyzed **600+ SKY130 physical-design runs**, comparing random search, Bayesian optimization, and failure-aware optimization while validating final layouts for timing, DRC, and LVS.

TidalGate | *C++, Python, Arduino Uno Q* | **DataHacks 2026 Winner**

- Integrated a Modulino IMU, Thermo module, and analog depth sensor onto an **Arduino Uno Q**, fusing 3D motion, temperature, and wave height data into a real-time tidal energy estimation pipeline.
- Wrote the data bridge between the embedded sensor layer and an on-device **Qualcomm-hosted Python ML model**, handling serialization and synchronization across heterogeneous platforms.

SKILLS

Programming: C, C++, Python, MATLAB, Verilog

Cleanroom Skills: Photolithography, Thin Film Measurement, DRIE, ALD, Mask Alignment, SPC

Hardware & Systems: Embedded Systems, PCB Design (KiCad), CATIA, Oscilloscopes

Development Tools: Git, VS Code, PlatformIO, Google Cloud Platform